

Lab 3 (Tutor Copy)**Learning Outcomes:**

1. Explain the purpose of exploratory data analysis (EDA)
2. Apply graphical and numerical techniques to discover patterns and anomalies.
3. Correlate data using descriptive statistics and basic visualization.
 - Able to create Scatter plot in TABLEAU
 - Able to determine if the correlation is positive, negative or none from the scatter plot
 - Able to use Linear Correlation node in KNIME to generate correlation matrix
 - Able to interpret the Pearson Linear Correlation Coefficient, r
 - Able to perform Exploratory Data Analysis

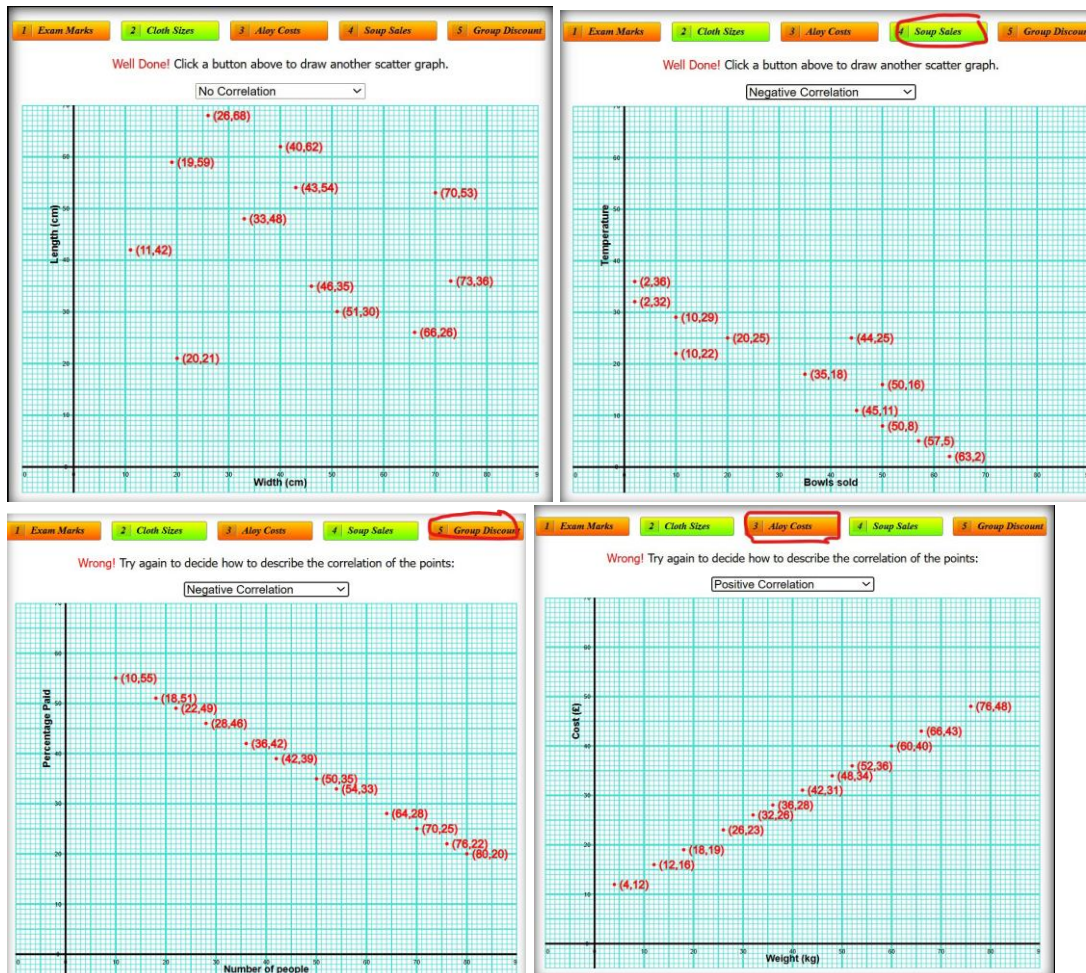
Activity 1: Concept of a scatter plot

Do the activity in this site:

[Plotting Scatter Graphs \(transum.org\)](https://transum.org)

Select any one scenario and plot the 2 variables on the same axes.

- I. Post a screen shot of the scatter plot here.
- II. Comment if there is any correlation between the 2 variables.



Tutor prompt/interaction:

Are these points plotted in any sequential order? => every dot is a data point, but they are not in collected in any sequential order.

Is this process tedious? Will it be good to automate this process?

Using this example to show data plotting is tedious, and analytics tool help to plot the data and we are able to observe pattern. It is not any magical program but simply plotting the points.

This example also shows plotting of 2 variables (as compared with boxplot and histogram).

What is the similarity between plotting the scatter plot and boxplot?

What is the difference between plotting the histogram and boxplot/scatter plot?

Boxplot and scatter plot show each data point on the graph.

Histograms summarize the count of data points within a "bin"/group.

There is "aggregation" operation when creating histogram.

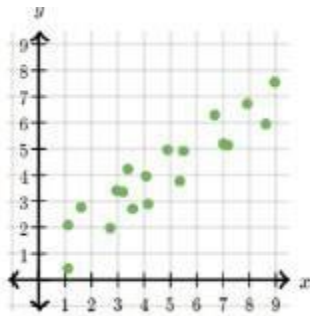
There is no aggregation operation when creating the boxplot and scatter plot.

Activity 2: Interpreting the scatter plot

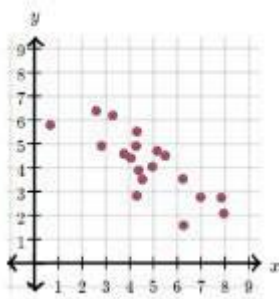
State if there is any linear correlation between the
2 variables, x and y

- positive correlation
- negative correlation or inverse correlation
- no correlated

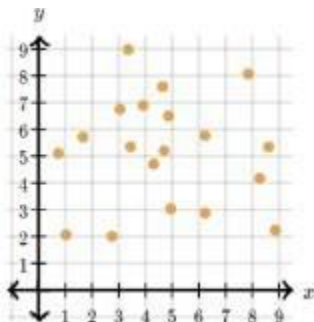
via the scatter plot below:



[Positive]



[Negative]



[No correlation]

Activity 3: Pearson Linear Correlation, r

The diagram illustrates the formula for the Sample Correlation Coefficient, r . It includes the following components and annotations:

- Sample Correlation Coefficient**: Points to the variable r in the formula.
- Summation: "Take The Sum Of"**: Points to the summation symbol Σ .
- Value of X**: Points to the variable x_i in the numerator.
- Mean of X Variable**: Points to the mean $\bar{x}$ in the numerator.
- Value of Y**: Points to the variable y_i in the numerator.
- Mean of Y Variable**: Points to the mean $\bar{y}$ in the numerator.
- Sum of the squared deviations for X**: Points to the term $\Sigma(x_i - \bar{x})^2$ in the denominator.
- Sum of the squared deviations for Y**: Points to the term $\Sigma(y_i - \bar{y})^2$ in the denominator.
- Square Root**: Points to the square root symbol $\sqrt{\quad}$ in the denominator.

$$r = \frac{\sum [(x_i - \bar{x})(y_i - \bar{y})]}{\sqrt{\sum (x_i - \bar{x})^2 * \sum (y_i - \bar{y})^2}}$$

This is the statistically method to determine the “strength” of the correlation. Strong or Weak

No need to memorize the formula. We will use KNIME program to compute for us. $r < 0$ (negative) $\Rightarrow$ negative correlation

$r > 0 \Rightarrow$ Positive correlation

$r > 0.7$ is relatively strong **positive** correlation.

$r < -0.7$ is relatively strong **negative** correlation. So, r value is between **-1** and **1**.

Exercise 1 – Create Scatter Plot in TABLEAU

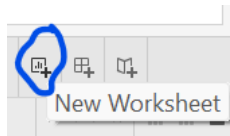
Data set: [CO2_data_Jan.xlsx](#)

You can either continue with your previous TABLEAU file (done in Lab 2)
or start a new TABLEAU workbook by reading the Excel file again.

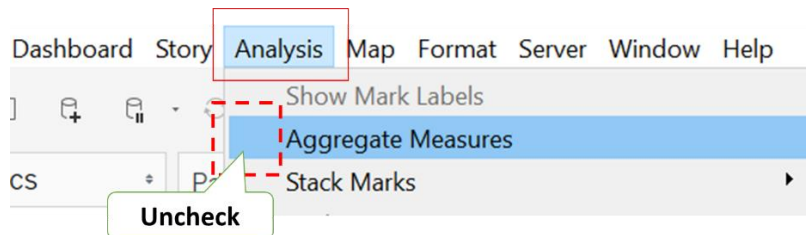
 CO2_Jan_workbook

Or start a new workbook.

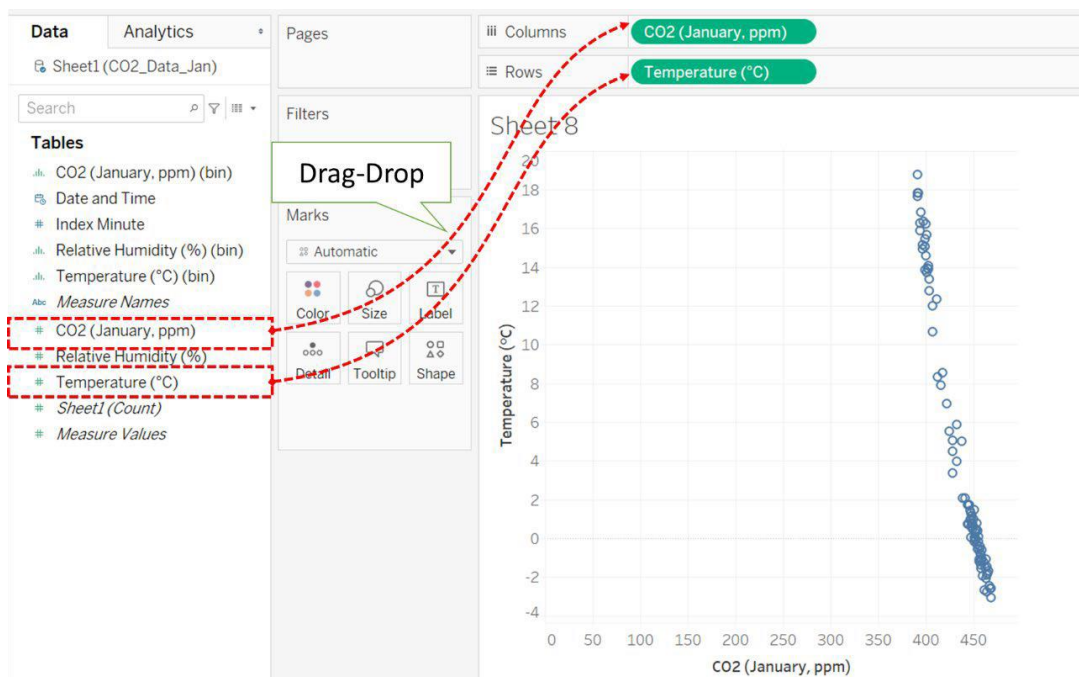
Start a new Sheet.



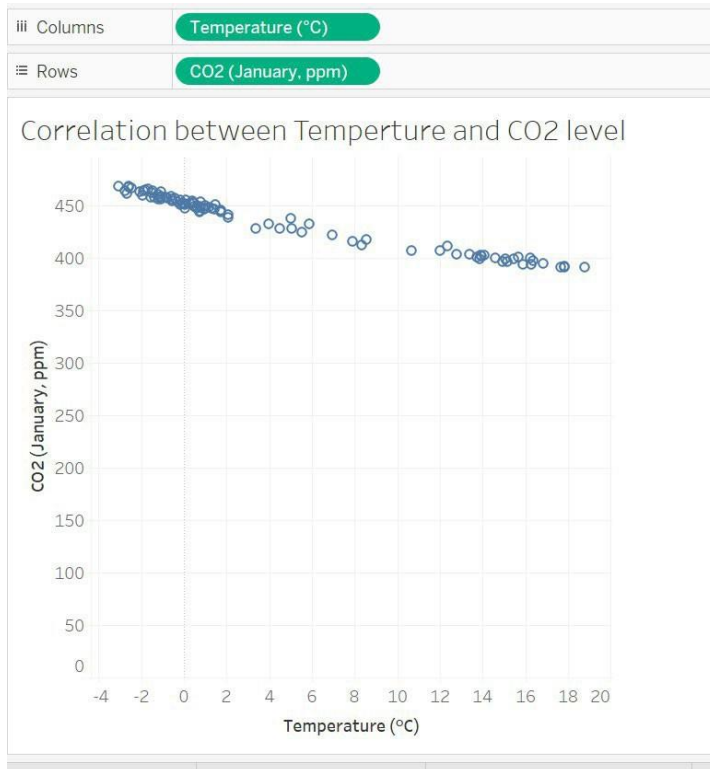
“Aggregate” option must be OFF for scatter plot, so that each data point is displayed (else TABLEAU will “sum” all values up to give a point only)



Let's analyse the relationship between CO2 level and
Temperature: Drag the variables to either columns/rows
shelves.

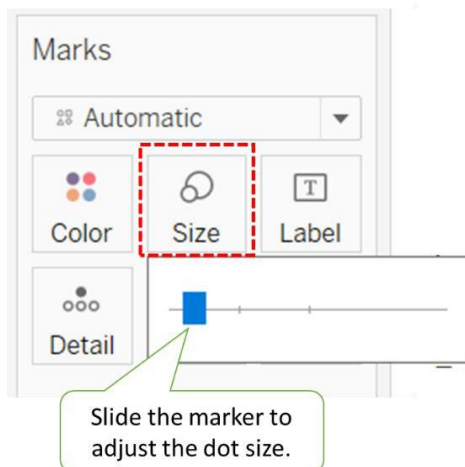


Alternatively, you may swap the columns and rows:



Both scatter plots show Strong negative correlation.

You may find the “dot” size too big. You can adjust the dot size smaller:





What is the correlation between CO2 and Temperature?

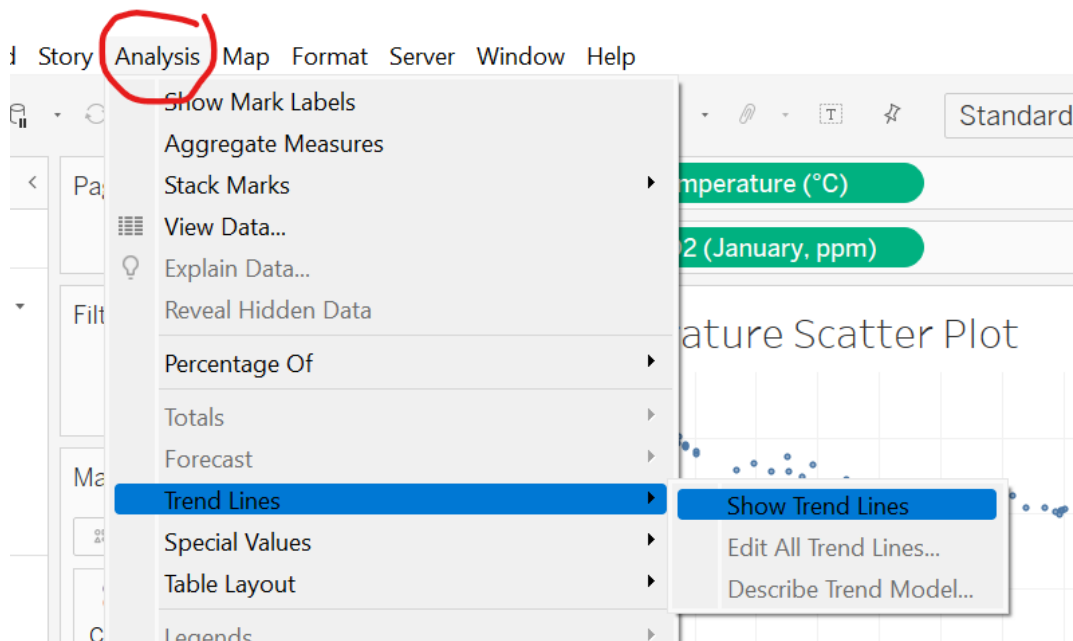
Strong Negative.

When temperature is high, CO2 level is low.

For a place with forest, or many trees, hotter day, like Day or Noon, the CO2 level will be lower, as photosynthesis take place.

Evening time, temperature usually lower, the tress are taking in the CO2, and so lower CO2 in the environment.

Can we fit a straight line to the scatter plot? Yes.



We observe a negative correlation between CO2 and temperature. Is this a Strong or Weak correlation?

Can we determine the correlation coefficient?

In fact, correlation is better observed in KNIME program. Let's save this workbook and get ready to use KNIME.

Getting Started with KNIME

In the next exercise, we will use KNIME to generate the correlation matrix.

Before we start, check that you have installed KNIME. We will go through the main features in KNIME first.

KNIME ANALYTICS PLATFORM 5.4

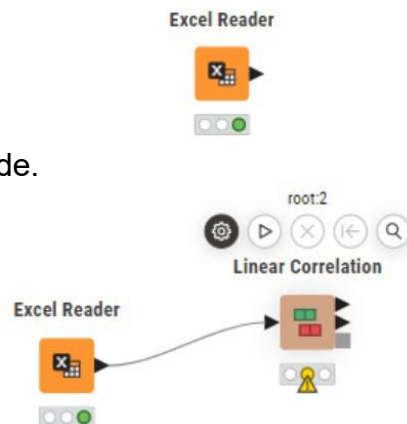
[Download | KNIME](#)

Exercise 2 – Find Linear Correlation Coefficient using KNIME

Data file: CO2_Data_April.xlsx

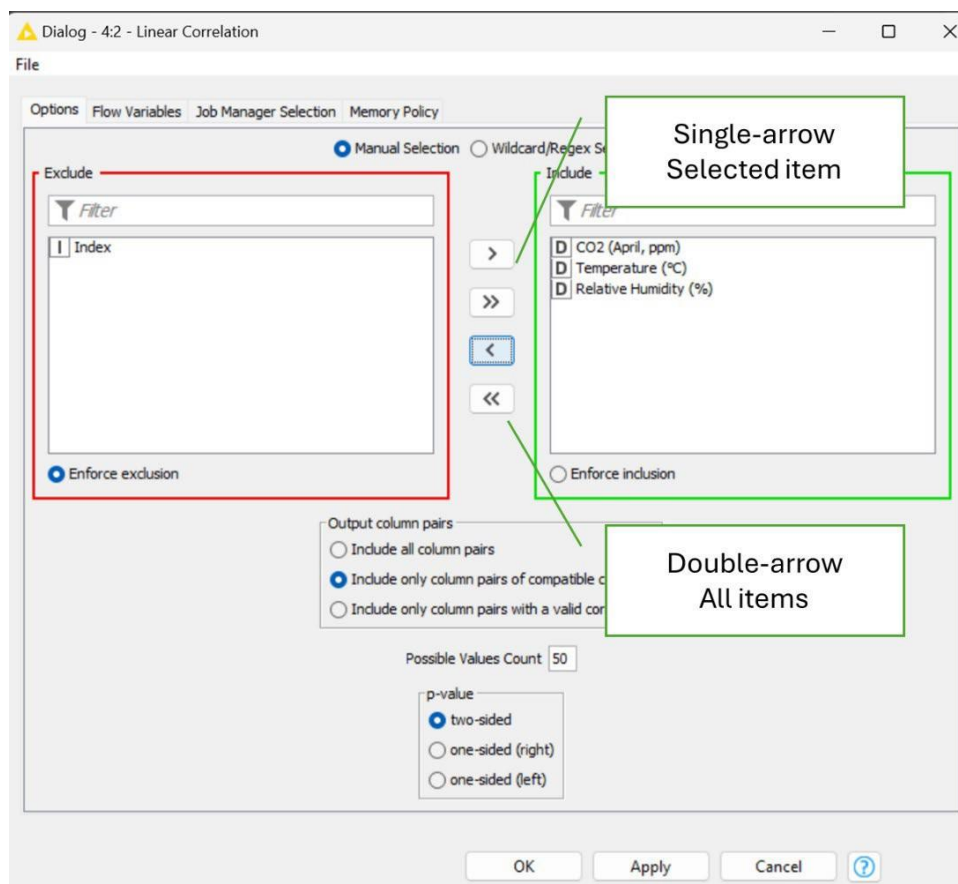
Create a new workflow in KNIME. Named the workflow: Lab3Ex2

Read in the data file using Excel Reader. Run the node (check the traffic light is green)
Add a Linear correlation node.

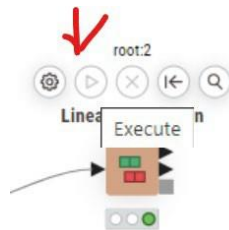


Add a Linear correlation node.

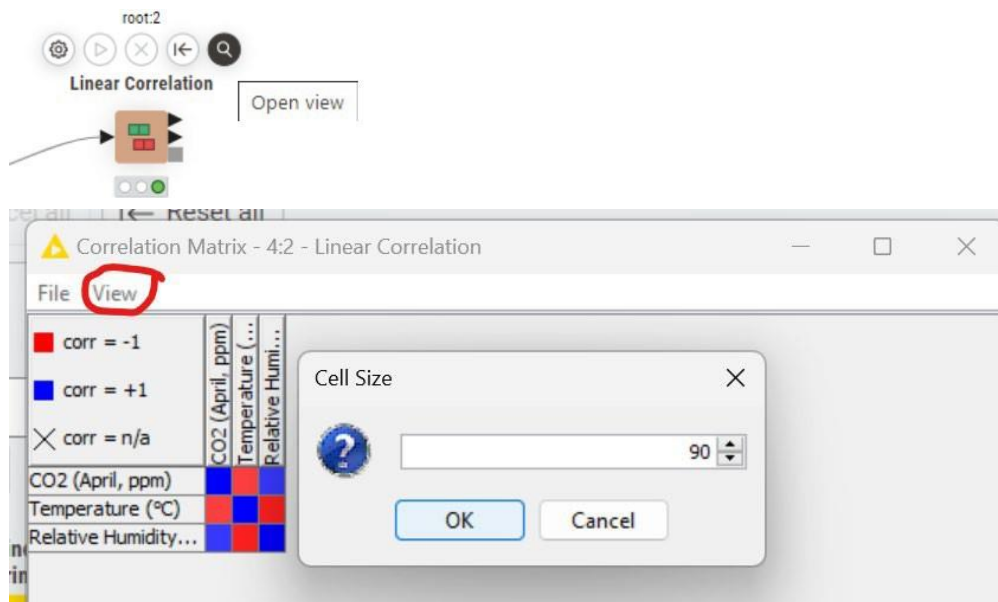
Configure the Linear Correlation node:



Run the Linear Correlation node. Check the traffic light is green.



Display the correlation Matrix:



Set the Cell size 90, to enlarge the view.

Interpret the Correlation Matrix:

Red – Negative Correlation
Blue – Positive Correlation



Activity:

What is the correlation between Temperature and CO2 level?

Temperature and CO2 are negatively correlated. They are inversely correlated. When temperature is high, CO2 level is low.

Which pair of variables have positive correlation?

Relative humidity and CO2 level is positively correlated. When CO2 is high, relative humidity is expected to be high.

Insights (additional interpretation. This is also an example what students can write for their project)

This could be due to photosynthesis happening in the day. The environment could be outdoor, with many trees.

At night, when temperature is low, CO2 level is high. The trees are giving out CO2 in the evening.

Temperature is also negatively correlated with Relative humidity. During the Day, temperature is high, the relative humidity is low. In the evening, temperature is low, the relative humidity is high.

Where can I find the strength of the correlation?

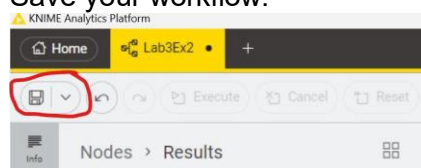
The Pearson Linear Correlation coefficient, r-value?

#	RowID	First column...	Second col...	Correlation...	p value	Degrees of ...
1	Row0	CO2 (April, ppm)	Temperature (°C)	-0.754	0	95
2	Row1	CO2 (April, ppm)	Relative Humidity	0.779	0	95
3	Row2	Temperature (°C)	Relative Humidity	-0.873	0	95

Select "Default" setting to view the complete decimal values:

#	RowID	First column...	Second col...	Correlation...	p value	Degrees of ...
1	Row0	CO2 (April, ppm)	Temperature (°C)	-0.75413485	0	95
2	Row1	CO2 (April, ppm)	Relative Humidity	0.779412551	0	95
3	Row2	Temperature (°C)	Relative Humidity	-0.87307613	0	95

Save your workflow.



Exercise 3 : Exploratory Data Analysis

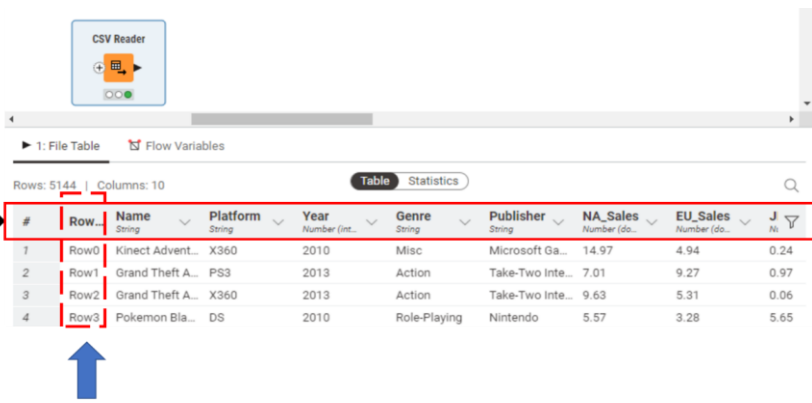
We will make use of a dataset containing video game sales data for various genres of video games on different gaming platforms. For more details, you can visit this link (<https://www.kaggle.com/gregorut/videogamesales>)

Data set: *vgsales_small.csv*

Data dictionary

Variable	Description
Rank	Ranking of overall sales
Name	Name of the game
Platform	Platform of the games release (eg. PC, PS4, etc)
Year	Year of the game's release
Genre	Genre of the game
Publisher	Publisher of the game
NA_Sales	Sales in North America (in millions)
EU_Sales	Sales in Europe (in millions)
JP_Sales	Sales in Japan (in millions)
Other_Sales	Sales in the rest of the world (in millions)
Global_Sales	Total worldwide sales

Understanding the data:



CSV Reader

► 1: File Table ✕ Flow Variables

Rows: 5144 | Columns: 10 Table Statistics

#	RowID	Name	Platform	Year	Genre	Publisher	NA_Sales	EU_Sales	J_Sales
1	Row0	Kinect Advent...	X360	2010	Misc	Microsoft Ga...	14.97	4.94	0.24
2	Row1	Grand Theft A...	PS3	2013	Action	Take-Two Inte...	7.01	9.27	0.97
3	Row2	Grand Theft A...	X360	2013	Action	Take-Two Inte...	9.63	5.31	0.06
4	Row3	Pokemon Bla...	DS	2010	Role-Playing	Nintendo	5.57	3.28	5.65

(a) Name this row

(b) Name this column

- General name for the first row is **Column Header**.
- General name for the first column is **RowID**.
- Each row represents a **video game title**.
- The columns are **variables**. (hint: observations or variables?)
- There are (how many?) **10** variables.
- There are (how many?) **5144** observations.

What insights can we gather from

this data? Question 1

Which region (NA_Sale, JP_Sale, EU_Sale, Other_Sale) has the best video game sales performance?

Hint: We can compare the Mean, Median, Max among these 4 regions.

Describe your findings. Post evidence of result (eg. Statistic)

Rows: 4 | Columns: 5

Name	50% Quantile (Median)	Mean ↓	Standard Devia...	Sum	Filter
NA_Sales	0.05	0.216	0.615	1,112.48	
EU_Sales	0.03	0.163	0.465	838.87	
JP_Sales	0	0.058	0.227	298.79	
Other_Sales	0.01	0.052	0.156	269.71	

North America sales are highest in term of sum, Median, and Mean. This region has the best sales performance.

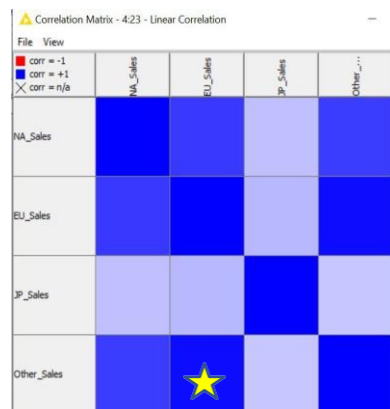
Question 2:

Are there any linear correlations among the sales in the 4 regions?

We need a Linear Correlation node to help us.

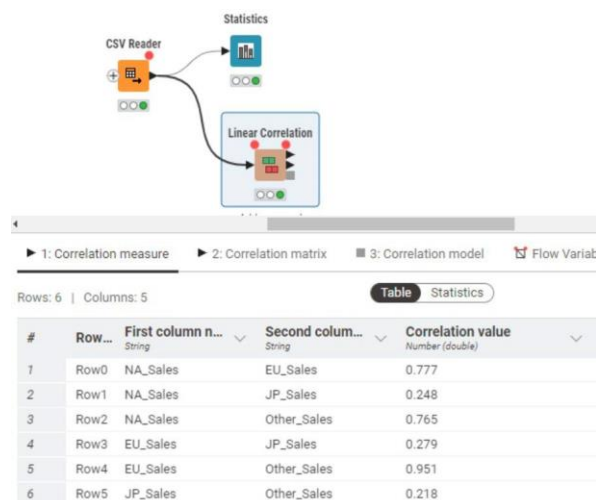
(a) Post your workflow, and the correlation matrix. Comment on the result.

You should select only numerical data variables.
Why “Global_Sales” was not analysed in this matrix?



Because Global_Sales is the sum of sales from these 4 regions. It is meaningless to find the correlation with Global_Sales.

Other_Sales is highly correlated to EU_Sales.



Correlation coefficient between “Other_Sales” and “EU_Sales” is 0.951.

Use TABLEAU to answer this question.

- (a) Create the scatter plot of *EU_Sales* and *Other_Sales* (using TABLEAU). Add a title. Post the result.



Exercise 4: Find Linear correlation (submit to LMS)

Data set: [SG24hrPSI.xlsx](#)

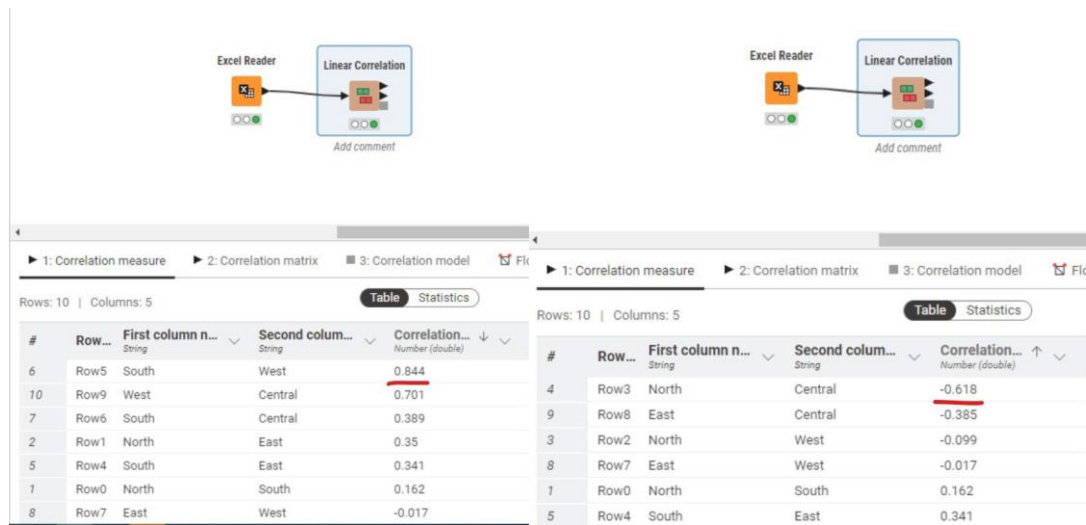
We have used this data set in week
2 before. Use KNIME for this
exercise.

Start a new **Workflow**.

Read the file using [Excel Reader](#).

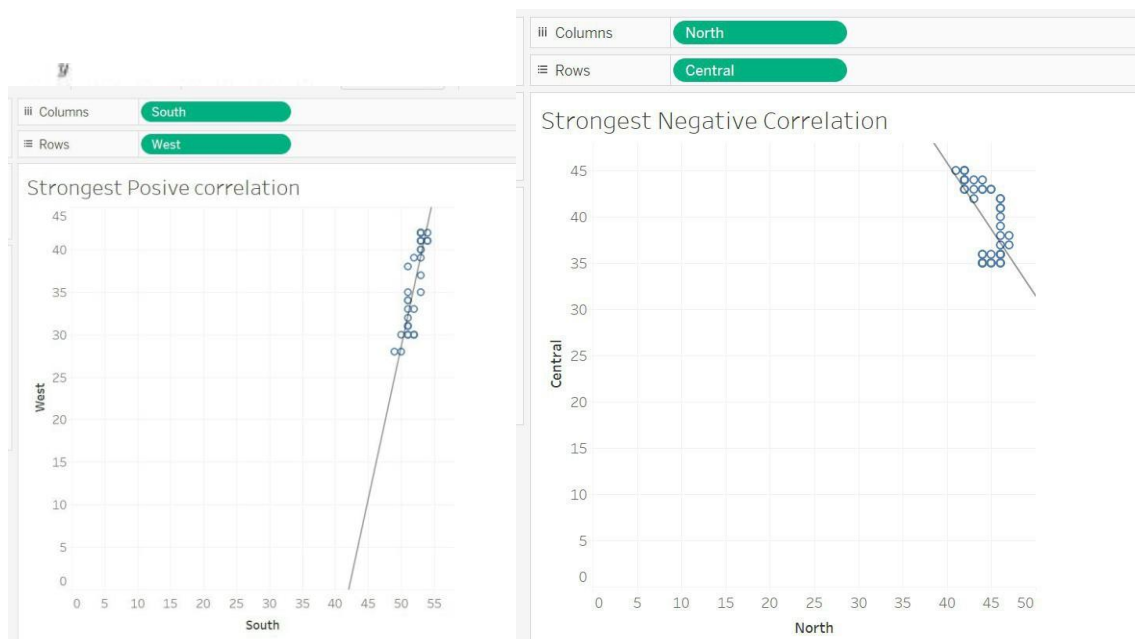
- (a) Post the correlation matrix (coloured cell) to analyse the correlation among the PSI from different regions (North, South, East, West and Central)





Use TABLEAU for these 2 questions:

- (b) Post the scatter plot of variables with highest positive correlation
- (c) Post the scatter plot of variables with highest negative correlation.



Both scatter plots show Strong negative correlation.
Yes,

This is additional reading material. It helps you to understand how to analyse the data and present the result. Keep this video in bookmark.

You can refer to this when you work on your project.

It explains the background of the data, then further into his investigative question, and show the various charts that help to answer his questions.

<https://youtu.be/puKodFOoIQw?si=936LFvi5mQ7SIVLY>